Outline of Fluid Mechanics

1 Fluid Properties

- Fluid as a Continuum
- Density (ρ) ; The density of a fluid is defined as its mass per unit volume.
- Specific Weight (γ);Specific weight is weight per unit volume.
- Specific Gravity (d); Specific gravity is the ratio of fluid density to the density of water at a certain temperature.
- Compressibility of Fluids (Bulk Modulus of Elasticity)
- Viscosity; viscosity is the property of fluid, which is a measure of its internal resistance to relative motion between adjacent layers.

$$\tau = \mu \frac{u}{y}$$

- Surface Tension
- Capillarity
- Vapor Pressure This partial pressure due to the vapors is known as *the vapor pressure.*

2 Fluid Statics

Pressure

- Hydrostatic Force on Immersed Plane Surfaces

$$F = \gamma h_G A \qquad h_p = \frac{I_G \sin^2\theta}{A h_G} + h_G$$

- Hydrostatic Force on Curved Areas

$$F_x = \gamma h_G A_x \qquad F_y = \gamma \forall \qquad F = \sqrt{F_x^2 + F_y^2}$$

- Pressure Variation in a Fluid at Rest

$$\begin{bmatrix} \frac{\partial p}{\partial x} = \rho X \\ \frac{\partial p}{\partial y} = \rho Y \\ \frac{\partial p}{\partial z} = \rho Z \end{bmatrix} , \qquad p = p_0 + \gamma h$$

- Relative Equilibrium
- Uniform Linear Acceleration
- Uniform Rotation about a Vertical Axis
- Stability of Floating Body

3 Fluid Kinematics

Analyzing of Fluid Mechanics

Eulerian and Lagrangian Methods

Fundamental Concepts of Fluid Flow

Fluid Types

Ideal Fluid: Inviscid fluids are fluids in which the fluid viscosity is assumed to be zero.

Viscous Fluid: Viscous fluids (real fluids) are fluids in which the fluid viscosity cannot be neglected.

Flow Types;

- Steady and Unsteady Flow; if the properties at a point in the flow field do not change with time, the fluid flow is called steady flow.

$$\frac{\partial E}{\partial t} = 0 \qquad E = E(x, y, z)$$

- Uniform and Non-uniform Flow; uniform flow is defined with respect to space at any instant of time

$$\frac{\partial E}{\partial s} = 0$$

Streamline

Pathlines

Stream Tube

System and Control Volume

Rotational and Irrotational Flows

Laminar and Turbulent Flows $\mathrm{Re} = \frac{V D}{\nu}$

Laminar flow is one in which the fluid flows in a well-ordered pattern whereby fluid layers are assumed to slide over one another

Re<2320 Laminar flow

In turbulent flow, the motion of individual fluid particles is highly erratic producing eddies and cross-currents.

Re>2320 Turbulent flow

Re≈2320 Transient flow

Fluid Acceleration

- convective or spatial acceleration
- local or temporal acceleration

Motion of a Fluid Element

- *Translation*
- *Linear Deformation*
- *Angular Deformation*
- *Rotation*

4 Fluid Dynamics

Forces

Surface Forces, Body Forces, Inertia Force, Elastic Force

Fundamental Principles

Conservation of mass gives *the continuity equation.*

$$\frac{\partial \rho}{\partial t}+\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}=0$$

Conservation of momentum gives *the dynamic equations or equations of motions*.

- Euler equations, which valid for inviscid fluid flow or frictionless flow.
- Navier-Stokes Equations valid for viscous fluid flow (Newtonian fluids)
- Reynolds Equations valid for turbulent flow

Conservation of energy (first law of thermodynamics) gives *the energy equation (*Bernoulli Eq. For ideal fluid).

$$\frac{V_1^2}{2g}+\frac{p_1}{\gamma}+z_1=\frac{V_2^2}{2g}+\frac{p_2}{\gamma}+z_2$$

Linear Impulse-Momentum Equation

$$\Sigma\vec{F}+\rho Q_1\overrightarrow{V_1}=\rho Q_2\overrightarrow{V_2}$$

Moment of Momentum Equation (Angular Momentum)

$$\Sigma M+\left(\rho Q_1\overrightarrow{V_1}\right)\vec{r}_1-\left(\rho Q_2\overrightarrow{V_2}\right)\vec{r}_2=0$$

Turbulent Shear Stress; $\bar{\tau}=-\rho\overline{u'v'}$

Total Shear Stress; $\tau=\mu\frac{du}{dy}+\left(-\rho\overline{u'v'}\right)$

Boundary Layer Concept ; For fluids having relatively small viscosity, the effect of internal friction in a fluid appreciable only in a narrow region surrounding the fluid boundaries

- Development of Boundary Layer
- Separation of Boundary Layer; and Wake

Hydrodynamics of Submerged Bodies

Hydrodynamic Forces; Drag Force, Lift Force

5 Introduction to Potential Flow

Rotation

$$\vec{\omega}=\frac{1}{2}\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ u & v & w\end{vmatrix}$$

If there is no rotation in a fluid flow $\vec{\omega}=0$ or $\text{curl}\vec{V}=\text{rot}\vec{V}=\nabla x\vec{V}=2\omega=0$

Potential and Stream Function

Bernoulli Equation in Potential Flow

$$\boxed{\frac{V_2^2}{2g}+\frac{p_2}{\rho}+y_2=\frac{V_1^2}{2g}+\frac{p_1}{\rho}+y_1=\text{constant}}$$

Flow Net

6 Dimensional Analysis and Modeling

The basis of dimensional analysis is to condense the number of separate variables involved in a particular type of physical system into a smaller number of non-dimensional groups of the variables.

Reynolds Number $=\frac{\text{Inertia Force}}{\text{Viscous Force}}=\frac{Vl}{\nu}$

Froude Number $=\frac{\text{Inertia Force}}{\text{Gravitational Force}}=\frac{V}{\sqrt{gl}}$

Weber Number $=\frac{\text{Inertia Force}}{\text{Surface Tension For.}}=\rho\frac{V^2 l}{\sigma}$

Euler Number $=\frac{\text{Pressure Force}}{\text{Inertia Force}}=\frac{p}{\rho V^2}$

Cauchy Number $=\frac{\text{Inertia Force}}{\text{Compressibility Force}}=\frac{\rho V^2}{E}$

Mach Number $=\sqrt{\text{Cauchy Number}}$

Strouhal Number $=\frac{\text{Inertia(local) Force}}{\text{Inertia(convective) Force}}=\frac{f\ell}{V}$

The Buckingham π Theorem